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SUPERSEDING
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MILITARY SPECIFICATION

OVENS, CRYSTAL,

GENERAL SPECIFICATION FOR

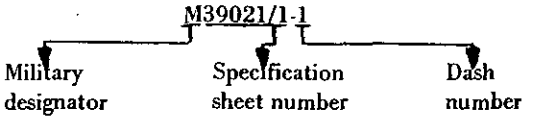
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for ovens designed to regulate the temperature of crystal units used in electronic equipment.

1.2 Classification.

1.2.1 *Military part number.* The military part number shall consist of the letter "M", the basic number of the specification sheet, and an assigned dash number (see 3.1) as shown in the following:



2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

QQ-C-585	Copper - Nickel - Zinc Alloy Plate, Sheet, Strip, and Bar (Copper Alloy Numbers 735, 745, 752, 762, 766 and 770).
QQ-S-571	Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-T-60	Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

MILITARY

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116	Preservation, Methods of.
MIL-S-6872	Soldering Process, General Specification For
MIL-S-12883	Sockets and Accessories For Plug-In Electronic Components.
MIL-C-45662	Calibration of Standards.

(See supplement 1 for list of applicable specification sheets.)

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STANDARD

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-1285	Marking of Electrical and Electronic Parts.
MIL-STD-1311	Test Methods For Electron Tubes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Ovens furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3)

3.3 First article inspection for ovens not covered by specification sheets. Unless otherwise specified (see 6.1.2), ovens furnished under this specification, not covered by specification sheets shall be a product which has been tested and passed the inspection specified in 4.5.

3.4 Material. Material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the ovens to meet the performance requirements of this specification. Acceptance or approval of any consitutent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5 Design and construction. Ovens shall be of the design, construction, and physical dimensions specified (see 3.1).

3.5.1 *Physical characteristics.* Physical characteristics (such as weight, finish, color, etc.,) shall be as specified (see 3.1).

3.5.2 *Heaters.* Ovens shall include built-in heaters and shall be capable of operation within performance requirements in any attitude or position. Heaters shall be wound in a manner to minimize inductance.

3.5.3 *Oven cavity.* The oven cavity shall satisfactorily accommodate the number of crystal units specified and shall provide for secure mechanical mounting of such crystal units. (See 3.1.)

3.5.3.1 *Grounding and circuit continuity.* The oven cavity shall provide means for a positive grounding of metal crystal holders to the external ground terminal of the oven. It shall also provide suitable electrical connection for crystal units. Where the oven has a metal enclosure, it shall be connected to the ground terminal. Interior to exterior electrical circuit continuity shall be as specified (see 3.1).

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3.5.4 *Accessibility.* Removal of crystal units from the oven shall be readily and easily accomplished, and shall be accomplished without use of special tools.

3.5.5 *Terminals.* Oven terminals shall have the mechanical configuration and provide electrical connections as specified (see 3.1).

3.5.5.1 *Identification.* Terminals shall be identified as specified (see 3.1).

3.5.5.2 *Pin alinement* (when specified). The pins in the base of the ovens shall freely and completely enter the specified pin-alinement test gage (see 3.1).

3.6 Insulation resistance. When ovens are tested as specified in 4.7.2, the insulation resistance shall be 100 megohms, minimum.

3.7 Dielectric withstanding voltage. When ovens are tested as specified in 4.7.3, there shall be no arcing, breakdown, or damage.

3.8 Capacitance. When ovens are tested as specified in 4.7.4, the capacitance between the terminals used for connection to the crystal unit shall be as specified (see 3.1).

3.9 Warmup power. When ovens are tested as specified in 4.7.5, the warmup power shall not exceed the value specified (see 3.1).

3.10 Warmup time. When ovens are tested as specified in 4.7.6, the warmup time (see 6.3.1) shall not exceed the value specified (see 3.1).

3.10.1 *Average operating power.* When ovens are tested as specified in 4.7.6.1, the average operating power of the oven at -55° C, unless otherwise specified (see 3.1), shall not exceed the value specified (see 3.1).

3.11 Oven cavity temperature.

3.11.1 *At room ambient.* When ovens are tested as specified in 4.7.7 and 4.7.7.1, the temperature limits after stabilization shall be as specified (see 3.1) for the following parameters:

(a) The recorded cavity temperature shall be within the specified base range.
(b) The actual mean operating temperature (determined from the initial recorded cavity temperature) shall be within the limits specified.
(c) Cyclic variations shall not exceed the value specified.

In tests made subsequent to the initial test, at any time following stabilization, the oven cavity temperature shall not exceed the value determined by the sum of the actual measured initial mean operating temperature of the specific unit plus one-half of the total permitted cyclic variations; nor shall this temperature at any time be less than the actual initial mean operating temperature minus one-half the permitted cyclic variation.

3.11.2 *At operating ambients.* When ovens are tested as specified in 4.7.7.2, the oven cavity temperature after stabilization shall be within the limits specified (see 3.1).

3.12 Vibration. When ovens are tested as specified in 4.7.8, the effects of vibration shall not cause contact flutter (see 6.2.3) which could affect the temperature control accuracy or warmup time or shorten life due to excessive contact operations. Random switching or abrupt change in cycling rate during or after the test shall be cause for rejection. If the temperature controlling element is a type which continuously draws power, any

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interruption or abrupt change in power consumption during the test shall be cause for rejection. The oven cavity temperature limits shall meet the requirements of 3.11.1. There shall be no resultant impairment or damage sufficient to cause failure of the holder or oven assembly.

3.13 Shock. When ovens are tested as specified in 4.7.9 there shall be no resultant impairment or damage sufficient to cause failure of the oven or displacement of the crystal holder, and the oven cavity temperature limits shall meet the requirements of 3.11.1.

3.14 Terminal strength.

3.14.1 *Terminal pull.* When ovens are tested as specified in 4.7.10.1, there shall be no loosening, fracturing, or rupturing of the terminals, or severing of wire leads; no other damage shall occur to the ovens.

3.14.2 *Wire-lead bend (applicable to ovens with wire leads).* When ovens are tested as specified in 4.7.10.2, there shall be no severing of the terminal.

3.14.3 *Twist (applicable to ovens with wire leads).* When ovens are tested as specified in 4.7.10.3, there shall be no severing of the terminal.

3.15 Thermal shock. When ovens are tested as specified in 4.7.11, there shall be no resultant impairment or damage sufficient to cause failure of the ovens, and the oven cavity temperature limits shall meet the requirements of 3.11.1.

3.16 Barometric pressure (reduced) (applicable to ovens for use above 10,000 feet altitude) (see 3.1). When ovens are tested as specified in 4.7.12, the dielectric withstanding voltage shall be as specified in 3.7. The oven cavity temperature limits shall meet the requirements of 3.11.1. There shall be no resultant impairment or damage sufficient to cause failure of the ovens.

3.17 Moisture resistance. When ovens are tested as specified in 4.7.13, there shall be no resultant impairment or damage sufficient to cause failure of the ovens, and the oven cavity temperature limits shall meet the requirements of 3.11.1.

3.18 Salt spray (corrosion). When ovens are tested as specified in 4.7.14, there shall be no warping, cracking, peeling, excessive corrosion, or in the case of plated metals, corrosion which has passed through the plating and exposed the base metal. After the test, the ovens shall be mechanically and electrically operable.

3.19 Explosion. When ovens are tested as specified in 4.7.15, the explosive mixture in the test chamber surrounding, and external to, the oven, shall not explode whether or not explosion occurs within the oven. The oven shall be electrically operable after the test.

3.20 Electromagnetic interference (EMI). When ovens are tested as specified in 4.7.16, the level of transient interference measured during warmup and during the current cycling mode shall not exceed 20 percent (peak voltage) of the applied line voltage unless otherwise specified (see 3.1).

3.21 Base pin solder depth (when specified). When ovens are tested as specified in 4.7.17, the depth of the solder contacts within the base pins shall not be less than the value specified in method 1111 of MIL-STD-1311.

3.22 Life. When ovens are tested as specified in 4.7.18, there shall be no resultant impairment or damage to cause failure of the ovens. At the specified test periods, the oven cavity temperature limits shall meet the requirements of 3.11.1, with maximum drift in mean operating temperature not to exceed the values specified.

3.23 Marking. The part number, heater voltage, oven cavity temperature, terminal identification, any special markings required, and date and source code shall be marked on the ovens in accordance with MIL-STD-1285 (see 3.1 and 6.1).

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3.24 **Workmanship.** Ovens including all parts and accessories shall be manufactured and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking, and plating. All dimensions and tolerances not specified shall be consistent with the best engineering practice. Where dimensions and tolerances affect the interchangeability, operation, or performance of the oven they shall be held or limited accordingly. All material shall be sound, of uniform quality and condition, and free from seams, cracks, and other defects which may adversely affect the strength, endurance, or wear of the part. Any material hammered, filed, or treated in any other manner to conceal defects therein, shall be subject to immediate rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Test equipment and inspection facilities.* Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the suppliers. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 **Classification of inspection.** The examination and testing of ovens shall be classified as follows:

(a) Qualification inspection.
(b) First article inspection for ovens not covered by specification sheets.
(c) Quality conformance inspection (see 4.4).

4.3 **Inspection conditions.**

4.3.1 *Inspection conditions.* Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 *Maximum overshoot.* Under any test conditions specified herein, the maximum temperature reached by the oven cavity shall not exceed the value specified (see 3.1).

4.4 **Qualification inspection.** Qualification inspection shall be performed at a laboratory acceptable to the Government (see 3.2 and 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 *Sample size.* Ten ovens, covered by a single specification sheet, shall be subjected to qualification inspection, except for grouping permitted in 4.4.2.

4.4.2 *Grouping for qualification.* Qualification obtained for ovens covered by a single specification sheet, in some instances, may constitute qualification for certain other ovens in the same group in accordance with table I.

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TABLE I. *Grouping for qualification.*

Group no.	MIL-O-39021 specification sheets
1	MIL-O-39021/1
2	MIL-O-39021/2, /4, /5, /6
3	MIL-O-39021/3, /8
4	MIL-O-39021/7
5	MIL-O-39021/9
6	MIL-O-39021/10, /11
7	MIL-O-39021/12
8	MIL-O-39021/13
9	MIL-O-39021/14
10	MIL-O-39021/15

4.4.3 *Inspection routine.* The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided equally into two groups and subjected to the inspection for their particular group.

4.4.4 *Failures.* One or more failures shall be cause for refusal to grant qualfication inspection.

TABLE II. *Qualification inspection.*

Examination or test	Requirement paragraph	Test method paragraph
Group I		
Visual and mechanical	3.1, 3.4, 3.5, 3.23 and 3.24	4.7.1
Insulation resistance	3.6	4.7.2
Dielectric withstanding voltage	3.7	4.7.3
Capacitance	3.8	4.7.4
Warmup power	3.9	4.7.5
Warmup time	3.10	4.7.6
Average operating power	3.10.1	4.7.6.1
Oven cavity temperature		
At room ambient	3.11.1	4.7.7.1
At operating ambient	3.11.2	4.7.7.2
Group II		
Vibration	3.12	4.7.8
Shock (when applicable)	3.13	4.7.9
Terminal strength	3.14	4.7.10
Thermal shock (when applicable)	3.15	4.7.11
Barometric pressure (reduced) (when applicable)	3.16	4.7.12
Moisture resistance (when applicable)	3.17	4.7.13
Insulation resistance	3.6	4.7.2
Salt spray (corrosion) (when applicable)	3.18	4.7.14
Explosion (when applicable)	3.19	4.7.15
Electromagnetic interference (when applicable)	3.20	4.7.16
Base pin solder depth (when applicable)	3.21	4.7.17
Group III		
Life	3.22	4.7.18

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4.4.5 *Retention of qualification.* To retain qualification, the supplier shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, (Group A), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) A summary of the results of tests performed for qualification verification inspection, (Group B), including the number and mode of failures. The summary shall include results of all qualification verification inspection tests performed and completed during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 2 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the products to testing in accordance with the qualification inspection requirements.

4.5 First article inspection of ovens not covered by specification sheets. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location suitable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production and shall be capable of meeting the inspections of 4.4 thru 4.4.5, inclusive. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.6 Quality conformance inspection.

4.6.1 *Inspection of product for delivery.* Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 *Inspection lot.* An inspection lot shall consist of all ovens with the same part number produced under essentially the same conditions and offered for inspection at one time.

4.6.1.2 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table III, and shall be made on the same set of sample units in the order shown.

TABLE III. *Group A inspection.*

Examination or test	Requirement paragraph	Method paragraph	AQL
Visual and mechanical	3.1, 3.4, 3.5, 3.23, and 3.24	4.7.1	1.0
Insulation resistance	3.6	4.7.2	1.5
Dielectric withstanding voltage	3.7	4.7.3	
Capacitance	3.8	4.7.4	
Warmup power	3.9	4.7.5	
Warmup time	3.10	4.7.6	
Average operating power	3.10.1	4.7.6.1	
Oven cavity temperature			
At room ambient	3.11.1	4.7.7.1	
At operating ambient	3.11.2	4.7.7.2	

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4.6.1.2.1 *Sampling plan.* Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality levels (AQL) shall be as specified in table III. Major defects shall be as defined in MIL-STD-105.

4.6.1.2.2 *Rejected lots.* If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.2.3 *Disposition of sample units.* Sample units which have passed all the group A inspection may be delivered on the contract or purchase order, if the lot is accepted.

4.6.2 *Qualification verification inspection.* Qualification verification inspection shall consist of group B. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.4), delivery of products which have passed group A shall not be delayed pending the results of these qualification verification inspections.

4.6.2.1 *Group B inspection.* Group B inspection shall consist of the tests specified in table IV, in the order shown. Group B inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

TABLE IV. *Group B inspection.*

Examination or test	Requirement paragraph	Test method paragraph
Subgroup 1		
Vibration	3.12	4.7.8
Shock (when applicable)	3.13	4.7.9
Terminal strength	3.14	4.7.10
Thermal shock (when applicable)	3.15	4.7.11
Barometric pressure (reduced) (when applicable)	3.16	4.7.12
Moisture resistance (when applicable)	3.17	4.7.13
Insulation resistance	3.6	4.7.2
Salt spray (corrosion) (when applicable)	3.18	4.7.14
Explosion (when applicable)	3.19	4.7.15
Electromagnetic interference (when applicable)	3.20	4.7.16
Base pin solder depth (when applicable)	3.21	4.7.17
Subgroup 2		
Life	3.22	4.7.18

4.6.2.1.1 *Sampling plan.* Ten sample units shall be selected from those ovens covered by the same specification sheet 12 months after the date of notification of qualification, and each 12 months thereafter. In the event that production has been suspended at the time when periodic inspection is due, the sample shall be selected from the first lot produced after production has been resumed. The sample shall be divided equally into two groups and subjected to the tests of subgroups 1 and 2 respectively of table IV.

4.6.2.1.2 *Failures.* If one or more sample units fail to pass group B inspection, the sample shall be considered to have failed.

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4.6.2.1.3 *Disposition of sample units.* Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.6.2.1.4 *Noncompliance.* If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 *Inspection of preparation for delivery.* Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 or the documents specified therein.

4.7 Methods of examination and test.

4.7.1 *Visual and mechanical examination.* Ovens shall be examined to verify that the materials. design, construction, physical dimensions, markings, and workmanship are in accordance with the applicable requirements. (See 3.1, 3.3, 3.5, 3.23 and 3.24.)

4.7.2 *Insulation resistance (see 3.6).* Ovens shall be tested in accordance with method 302 of MIL-STD-202. The following details shall apply:

(a) Test condition - B.
(b) Points of measurement - Between all grounded terminals connected together, and all ungrounded terminals connected together, except that for ovens using grounded power sources they shall be as specified (see 3.1).
(c) Electrification time - One minute.

4.7.3 *Dielectric withstanding voltage (see 3.7).* Ovens shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude of test voltage - Unless otherwise specified (see 3.1), 500 volts rms per second for a period of 60 seconds for qualification and quality conformance testing.
(b) Nature of potential - AC.
(c) Points of application of test voltage - As specified in 4.7.2(b).

Following this test, ovens shall be examined for evidence of arcing, breakdown, and damage.

4.7.4 *Capacitance (see 3.8).* Ovens shall be tested in accordance with method 305 of MIL-STD-202. The following detail and exception shall apply:

(a) Test frequency - 100 kHz or 1 MHz.
(b) Points of measurement - Between the terminals used for connection to the crystal unit.

4.7.5 *Warmup power (see 3.9).* Ovens, nonoperating, shall be subjected to an ambient temperature of -55° C, unless otherwise specified (see 3.1), for a period of 3 hours. While the ovens remain in this ambient temperature, rated voltage (see 3.1) shall be applied and the warmup power to the oven heaters shall be measured immediately.

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4.7.6 *Warmup time (see 3.10).* Ovens, nonoperating, shall be subjected to an ambient temperature of -55° C, unless otherwise specified (see 3.1), for a minimum period of 3 hours. While the ovens remain in this ambient temperature, minimum rated voltage (see 3.1) shall be applied and the elapsed time shall be measured from the instant of application of voltage to the time when the oven cavity temperature comes to and remains within the limits specified (see 3.1).

4.7.6.1 *Average operating power (see 3.10.1).* The measurement for average operating power shall be made at -55° C ambient extreme, unless otherwise specified (see 3.1), after the specified warmup time (see 3.1). Where cyclic thermostatic control is used, this measurement is determined by measuring the peak power while the operating thermostat is closed, the ON time, and the OFF time. The average power after stabilization is then calculated using the formula:

$$\frac{\text{ON time x peak power}}{\text{ON time + OFF time}} = \text{Average operating power}$$

4.7.7 *Oven cavity temperature (see 3.11).* The temperature-sensing standard for this test shall be a digital thermometer which has the following characteristics.

Accuracy (digital readout and probe) .	±0.25° C
Accuracy (probe)	±0.1° C
Resolution	0.1° C
Repeatability	0.1° C (-55 to 105° C)

The temperature sensing probe must be certified, traceable to National Bureau of Standards with regards to absolute temperature. Data to substantiate this accuracy shall be available at -55° , 0°, 23°, and 100° C to a resolution of 0.1° C.

4.7.7.1 *At room ambient (see 3.11.1).* After stabilization, the oven cavity temperature shall be recorded at room ambient temperature.

4.7.7.2 *At operating ambients (see 3.11.2).* The oven cavity temperature shall be continuously monitored throughout the test. The ovens shall be maintained at an ambient temperature of -55° C, unless otherwise specified (see 3.1), for a period of three hours prior to the temperature run. Following recorded measurement verifying stabilization at -55° C, unless otherwise specified (see 3.1), the ambient temperature shall be increased to +25° C until stabilization is achieved. The ambient temperature shall then be increased to the upper extreme (see 3.1) until stabilization is achieved. Unless otherwise specified (see 3.1), the actual measured ambient at each of the three levels shall be within ±1° C of the specified temperatures. The ambient air temperature shall be measured at a point not closer than three unobstructed inches from either the oven being tested or the interior wall of the test chamber. The air in the test chamber shall be under constant forced circulation at the rate of 50 ±25 feet per minute.

4.7.8 *Vibration (see 3.12).* Ovens shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting of specimens - Rigidly mount to a metal panel, unless a mounting device is furnished as a component part of the oven (see 3.1). For cylindrical ovens the axis formed by pin 1 and the center of the oven base shall be one of the radial axes of vibration.
(b) Electrical-load conditions - Normal heater connections shall be made.
(c) Test condition - A.
(d) Measurements during test - Contact flutter shall be observed by an oscilloscope which will show the normal cyclic contact operation when the oven cavity temperature is within the specified

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limits. Contact flutter may be observed by the effect on normal steady heater current during warmup (vibratory openings); by the effect on warmup time checked at the vibration frequency where flutter is most pronounced; by a change in the normal cyclic opening and closing observed without vibration when the oven cavity temperature is being controlled with room ambient. Oven power shall be monitored. If the temperature controlling element is a cyclic device, power consumption shall continue to cycle at the same rate as in the absence of vibration. If any spurious contact operations are observed due to vibration, the supplier shall be required to prove that these will have no deleterious effect on the functional requirements of the unit, such as life, warmup time, temperature control or radio interference;.

(e) Measurement after test - Ovens shall be tested as specified in 4.7.7.1 and 4.7.7.2.

Following this test, ovens shall be examined for evidence of impairment or damage which would cause failure of the holder or oven assembly.

4.7.9 *Shock (see 3.13).* Unless otherwise specified (see 3.1), ovens shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting method - As specified in 4.7.8(a).
(b) Test condition - I.
(c) Number of blows - Ovens shall be subjected to two blows in each of six directions along the three major axes, for a total of twelve blows.
(d) Measurements after shock - Ovens shall be tested as specified in 4.7.7 and 4.7.7.1.

Following this test, ovens shall be examined for impairment or damage which would cause failure of the oven or displacement of the crystal holder.

4.7.10 *Terminal strength.*

4.7.10.1 *Terminal pull (see 3.14.1).* Unless otherwise specified (see 3.1), ovens shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.
(b) Applied force - 4-1/2 lbs for pin terminals; 2 lbs for wire lead terminals (applied to each terminal).

4.7.10.2 *Wire lead bend (applicable to ovens with wire leads) (see 3.1 and 3.14.2).* Ovens shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply:

(a) Test condition - C.
(b) Applied load - 1 pound.

4.7.10.3 *Twist (applicable to ovens with wire leads) (see 3.1 and 3.14.3).* Ovens shall be tested in accordance with method 211, test condition D.

4.7.11 *Thermal shock (see 3.15).* Unless otherwise specified (see 3.1), ovens shall be tested in accordance with method 107 of MIL-STD-202. The following detail and exception shall apply:

(a) Test condition - A.
(b) Measurements after cycling - Ovens shall be tested as specified in 4.7.7 and 4.7.7.1.

Following the test, ovens shall be examined for impairment or damage which would cause failure of the ovens.

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4.7.12 *Barometric pressure (reduced) (applicable to ovens for use above 10,000 feet altitude) (see 3.1) (see 3.16).* Ovens shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - By normal mounting means in the normal position and attitude.
(b) Test condition - C.
(c) Test during subjection to reduced pressure - Dielectric withstanding voltage shall be measured as specified in 4.7.3, except that the magnitude of the test voltage shall be 200 volts, rms for a period of 1 minute.
(d) Test after subjection to reduced pressure - Ovens shall be tested as specified in 4.7.7 and 4.7.7.2.
(e) Exposure time prior to measurements - One hour.

Following this test, ovens shall be examined for impairment or damage which would cause failure of the ovens.

4.7.13 *Moisture resistance (see 3.17).* Unless otherwise specified (see 3.1), ovens shall be tested in accordance with method 106 of MIL-STD-202. The following details shall apply:

(a) Measurements after conditioning - Ovens shall be tested as specified in 4.7.7 and 4.7.7.1.
(b) Test condition - B.
(c) Final measurements - Ovens shall be tested as specified in 4.7.7 and 4.7.7.1 (after drying period).

Following this test, ovens shall be examined for impairment or damage which would cause failure of the ovens.

4.7.14 *Salt spray (corrosion) (see 3.18).* Unless otherwise specified (see 3.1), ovens shall be tested in accordance with method 101 of MIL-STD-202. The following details shall apply:

(a) Test condition - B.
(b) Measurements after exposure - Ovens shall be tested as specified in 4.7.7 and 4.7.7.1.

Following this test, ovens shall be examined for evidence of excessive corrosion or corrosion that would interfere with the electrical or mechanical performance of the ovens.

4.7.15 *Explosion (see 3.19).* Unless otherwise specified (see 3.1), ovens shall be tested in accordance with method 109 of MIL-STD-202. The following detail shall apply:

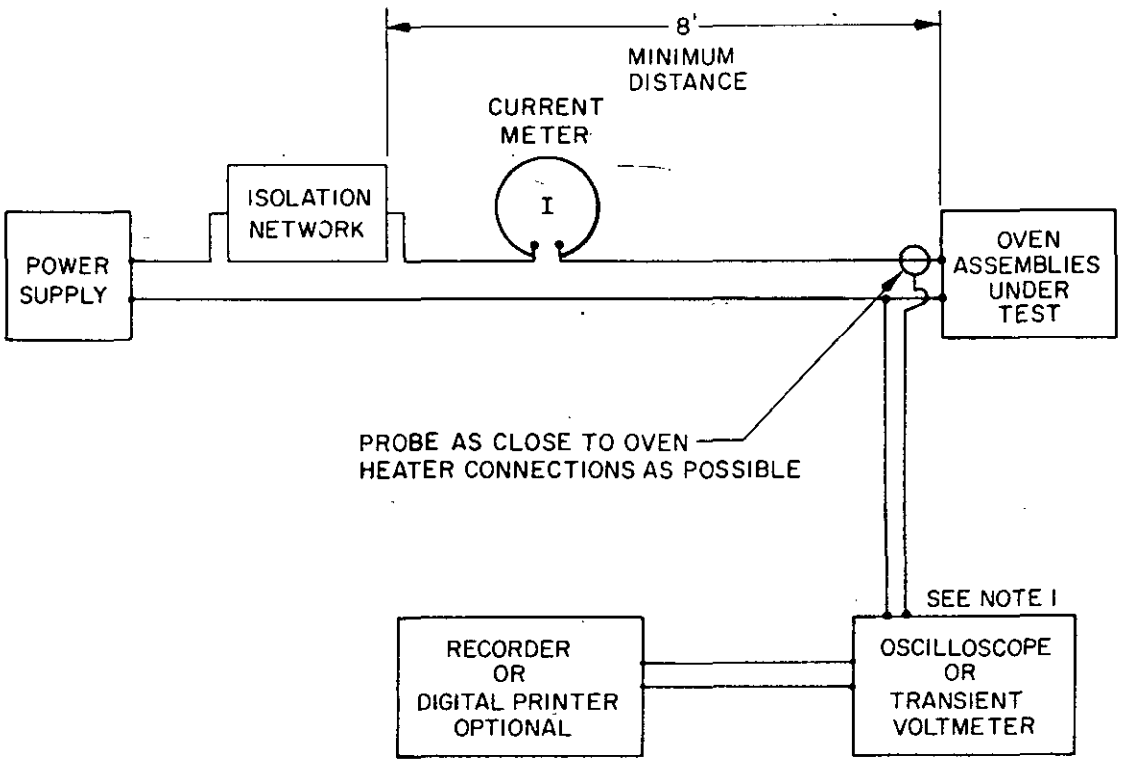
(a) Mechanical and electrical load - Ovens shall be operated at maximum rated dc inductive current.

4.7.16 *Electromagnetic interference (EMI) (see 3.20).* Unless otherwise specified (see 3.1), the oven assembly shall be tested for transient interference at room ambient temperature in the test circuit as shown in figure 1. Input power to the oven heater element shall be at maximum voltage and shall not exceed the specified limits. The oven shall be monitored for noise continuously from initiation of input power to the oven heater element and during the time in which the oven is in its current cycling mode as indicated by the current meter in the power line. During the current cycling mode monitoring for noise shall be continued for a time sufficient to indicate that no significant increase in noise is generated by the oven under test. The highest reading obtained shall be recorded. Means to accelerate the number of cycles during the current cycling mode after warm up may be applied by providing a low environmental temperature to the oven. The low temperature environment shall not exceed the -55° C limit, unless otherwise specified (see 3.1).

4.7.17 *Base pin solder depth (see 3.21).* Unless otherwise specified (see 3.1), ovens shall be tested in accordance with method 1111 of MIL-STD-1311.

4.7.18 *Life (see 3.22).* Ovens shall be continuously cycled "on" and "off" for a total of 1,000 hours. One cycle shall consist of 1 hour "on" (energy supplied) and 1 hour "off". Oven temperature shall be measured for a

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NOTES:

1. Oscilloscope or transient voltmeter shall be of the memory storage type and be capable of detecting and measuring:

Magnitude:	±1 volt to ±1000 volts
Width:	50 nanoseconds to 100 milliseconds
PRF:	Single shot to 500 PPS
Input Impedance:	100k ohms min

2. EMI isolation network can be a low pass filter providing 40 to 60 dB stop band attenuation in the test circuit.
3. Power supply shall have low internal impedance.
4. Test shall be performed in a shielded room.
5. Power supply leads may be lengthened for additional inductance.

FIGURE 1. Electromagnetic interference.

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period of 24 hours at the completion of 500 hours of cycling; and for a period of 24 hours after completion of 1,000 hours of cycling. Ovens shall be tested as specified in 4.7.7 and 4.7.7.1. After the test, ovens shall be examined for any impairment or damage which would cause failure of the ovens.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* Ovens shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 *Drying.* Ovens shall be dried in accordance with MIL-P-116.

5.1.1.3 *Preservative application.* None required.

5.1.1.4 *Unit packaging.* Ovens shall be individually packaged in accordance with MIL-P-116, submethod IIc insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Each unit package shall be placed in a supplementary container conforming to PPP-B-566 or PPP-B-676.

5.1.1.5 *Intermediate packaging.* Not required.

5.1.2 *Level C.* Cleaned and dried ovens shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 *Level A.* The packaged ovens shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

5.2.2 *Level B.* The packaged ovens shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto. For Army procurement, fiberboard containers shall be class weather-resistant as specified in level A.

5.2.3 *Level C.* The packaged ovens shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 *Unitized loads.* Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 *Level A.* Ovens, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

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5.2.4.2 *Level B.* Ovens, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic. For Army procurement, caps shall be weather resistant as specified in level A.

5.2.4.3 *Level C.* Ovens, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, supplementary and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 *Exterior containers.* Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 *Navy procurements.* For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

5.4.3 Indirect shipments. The preservation and packaging, packing and marking specified in section 5 are intended to apply only to direct purchases by or direct shipments to the Government. However, all or part of section 5 may be cited in purchase documents between contractors and suppliers.

6. NOTES

6.1 Ordering data.

6.1.1 *For ovens covered by specification sheets.* Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete part number.
(c) Any special marking required (see 3.23).
(d) Levels of preservation and packaging and packing (see section 5).

6.1.2 *For ovens not covered by specification sheets.* The procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Manufacturer's part number.
(c) Marking requirements (see 3.21).
(d) Levels of preservation and packaging and packing (see section 5).
(e) Applicable qualification or first article requirements if other than those specified herein (e.g. number of sample units, submission of data, test routine, and other information as may be required).

6.2 Definitions.

6.2.1 *Warmup time.* The period of elapsed time between the application of energy to the heaters, and the instant when the oven cavity temperature comes to and stays within the specified limits.

6.2.2 *Ambient temperature.* The temperature of the air surrounding the oven, and measured as specified in 4.7.7. Room ambient temperature is in accordance with the general requirements of MIL-STD-202. Operating

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ambient is the ambient temperature range that the oven is designed to encounter in service, and over which the oven must maintain control of the oven cavity temperature.

6.2.3 *Contact flutter.* Spurious contact operation which may cause contact opening during the warmup period, or cause periodic closure when the chamber is above the upper limit, or result in a marked change in the normal cyclic opening and closing which occurs with the specific unit during the normal control without vibration.

6.2.4 *Proportional oven.* Proportional control ovens apply power in proportion to the temperature differential, thermal losses, and device power dissipation inside cavity in order to maintain a constant cavity temperature.

6.3 **Qualification.** With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the U.S. Army Electronics Command, Fort Monmouth, New Jersey 07703; however, information pertaining to qualification of products may be obtained from Defense Electronics Supply Center (DESC-EQ), Dayton, Ohio 45401.

6.4 **First article inspection.** Information pertaining to first article inspection of ovens not covered by specification sheets should be obtained from the procuring activity for the specific contracts involved.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - MI, MU, EL, SM
Navy - OS
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, OS, SH
Air Force - 14, 19, 84

Preparing activity:
Army - EL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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